

DATA DRIVEN INSIGHTS USING PYTHON – PROJECT

Austo Motor Company: Data-Driven Analysis of Marketing Strategy

Background:

Austo Motor Company is a leading automobile manufacturer specializing in SUVs, Sedans, and Hatchbacks. In a recent board meeting, concerns were raised regarding the efficiency of the company's marketing campaigns and its alignment with customer preferences. To address this, the board engaged the Data Science team to analyse customer and purchase data in order to identify patterns, preferences, and potential opportunities to improve marketing effectiveness and sales strategy.

Objective:

- To assess the influence of profession, gender, age, and household income on customer's vehicle choice and price.
- To evaluate how loans and dual-income households impact spending patterns.
- To provide actionable insights for targeted marketing, product positioning, and sales optimization.

Data Description:

The dataset contained 1,581 customer records with the following 14 fields:

Column	Description
Age	Age of the individual (years)
Gender	Gender (Male/Female)
Profession	Occupation (Salaried/Business)
Marital_status	Marital status (Married/Single)
Education	Graduate / Post Graduate
No_of_Dependents	Number of financial dependents
Personal_loan	Whether the customer has a personal loan (Yes/No)
House_loan	Whether the customer has a housing loan (Yes/No)
Partner_working	Whether the partner is employed (Yes/No)
Salary	Individual salary
Partner_salary	Partner's salary
Total_salary	Combined household salary
Price	Price of the purchased vehicle
Make	Vehicle type (SUV, Sedan, Hatchback)

- The dataset consisted of 1,581 observations and 14 variables, with minimal missing data concentrated in the Gender and Partner_salary columns.
- All numerical and categorical variables were stored using appropriate data types, indicating good overall data quality.

#	Column	Non-Null Count	Dtype
0	Age	1581 non-null	int64
1	Gender	1528 non-null	object
2	Profession	1581 non-null	object
3	Marital_status	1581 non-null	object
4	Education	1581 non-null	object
5	No_of_Dependents	1581 non-null	int64
6	Personal_loan	1581 non-null	object
7	House_loan	1581 non-null	object
8	Partner_working	1581 non-null	object
9	Salary	1581 non-null	int64
10	Partner_salary	1475 non-null	float64
11	Total_salary	1581 non-null	int64
12	Price	1581 non-null	int64
13	Make	1581 non-null	object

Statistical Summary of the dataset:

	count	mean	std	min	25%	50%	75%	max
Age	1581.0	31.922201	8.425978	22.0	25.0	29.0	38.0	54.0
No_of_Dependents	1581.0	2.457938	0.943483	0.0	2.0	2.0	3.0	4.0
Salary	1581.0	60392.220114	14674.825044	30000.0	51900.0	59500.0	71800.0	99300.0
Partner_salary	1475.0	20225.559322	19573.149277	0.0	0.0	25600.0	38300.0	80500.0
Total_salary	1581.0	79625.996205	25545.857768	30000.0	60500.0	78000.0	95900.0	171000.0
Price	1581.0	35597.722960	13633.636545	18000.0	25000.0	31000.0	47000.0	70000.0

- Majority of the customers were in the age group between 25 and 38 with an average age of 32.
- The stats indicated that there could be many unemployed partners.
- The price ranges were distributed from 18000 dollars to 70000 dollars indicating a wide distribution.

Data Cleaning:

- The dataset contained missing values in two key columns: Gender and Partner_salary.

Gender:

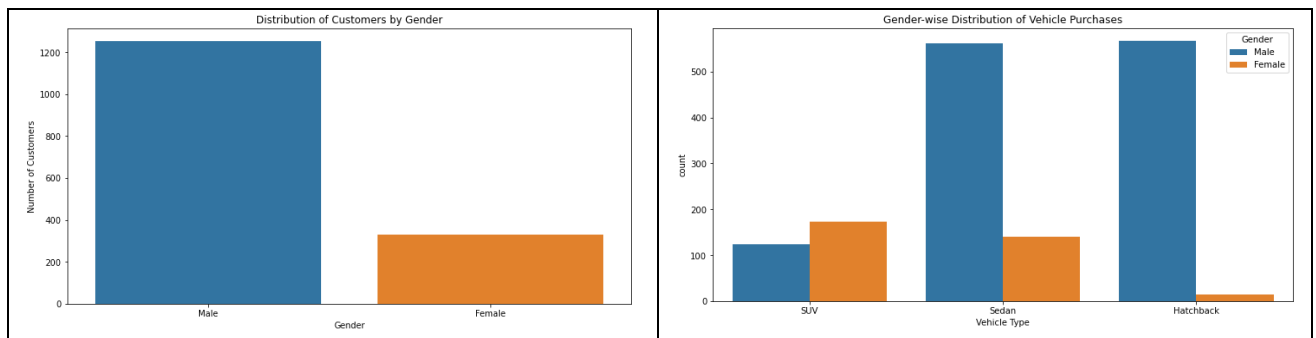
- A small number of records had missing or inconsistent entries (e.g., Femal, Femle).
- These inconsistencies were corrected, and missing values were replaced with the mode of the column (Male) to maintain data consistency.

Partner_salary:

- Some records had missing partner salary information, primarily for single individuals or incomplete submissions.
- Missing values were replaced with the median of the column to avoid skewing the dataset, as salary distributions can be influenced by extreme values.
- After these imputations, all columns were free of missing values, ensuring a complete dataset for reliable analysis.
- Additional cleaning steps included standardizing categorical values and verifying there were no duplicate records, ensuring high-quality and consistent data for subsequent analysis.

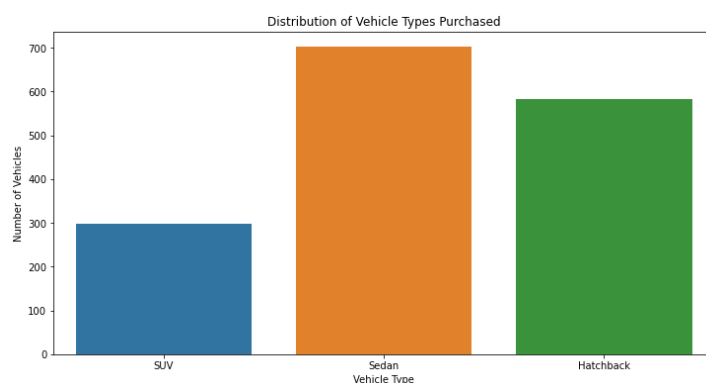
Univariate Analysis:

Distribution of Customers by Gender:



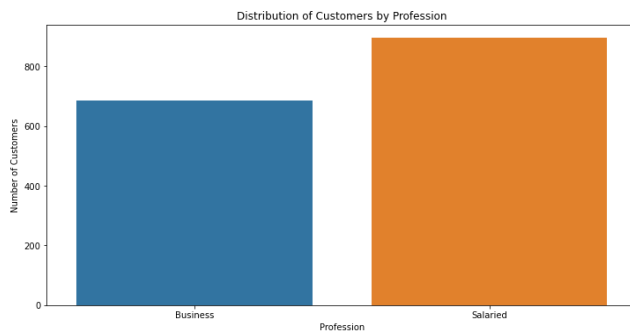
- Gender-wise analysis reveals a clear differentiation in vehicle preferences.
- Female customers show a strong preference for SUVs, with over 52% of purchases falling in this category, while Hatchbacks account for less than 5% of their purchases.
- In contrast, male customers exhibit a more evenly distributed preference between Hatchbacks and Sedans, each contributing approximately 45% of purchases, with SUVs forming a relatively small share.

Distribution of Vehicle Types Purchased:



- The univariate analysis of vehicle types reveals that the overall demand is skewed toward Sedans and Hatchbacks, which together account for over 80% of total purchases.
- Sedans dominate customer demand, accounting for approximately 44% of total purchases.
- Hatchbacks follow closely with nearly 37%, highlighting a strong market for compact and budget-friendly vehicles.
- SUVs, in contrast, represent a smaller segment at around 19%, suggesting that while present, they cater to a more niche customer base.

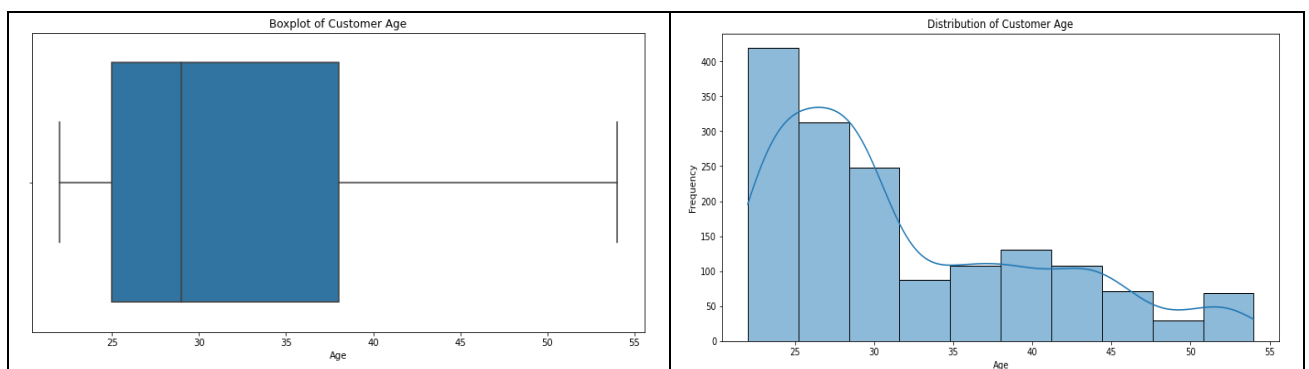
Distribution of Customers by Profession:



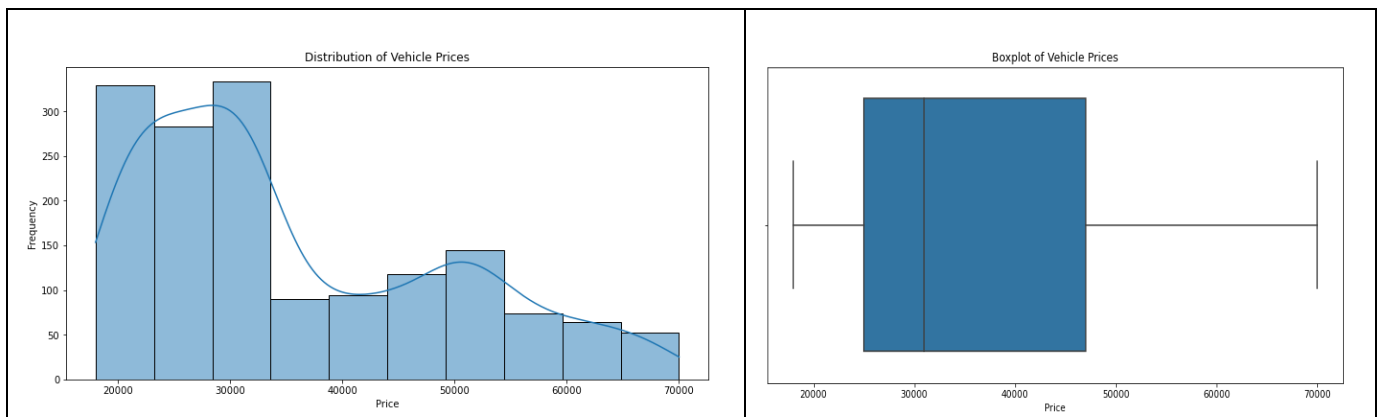
- The univariate analysis of profession shows a relatively balanced distribution of customers, with salaried individuals accounting for approximately 57% of purchases and business professionals contributing about 43%.
- This indicates that Austo Motor Company serves a diverse professional base, rather than dependence on a single professional segment.

Distribution of Customer Age:

- The univariate analysis of customer age reveals that Austo Motor Company primarily serves a young-to-mid aged customer base.
- With a median age of 29 years and a mean age of approximately 32 years, the distribution is slightly right-skewed, indicating fewer older customers.

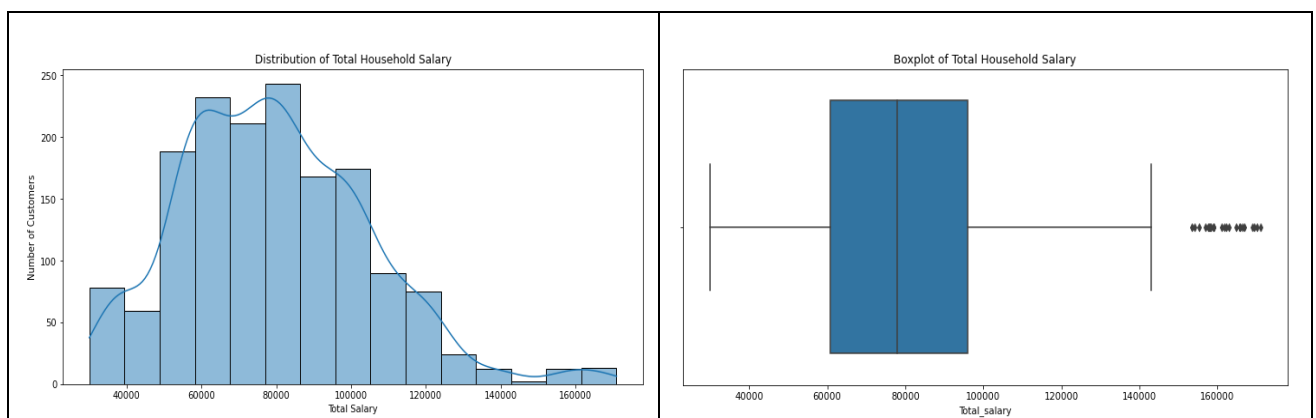


Distribution of Vehicle Prices:



- The univariate analysis of vehicle prices reveals a wide but right-skewed distribution. While prices range from 18,000 to 70,000, most customers operate within a lower-to-mid price band.
- The interquartile range shows 25% of purchases are priced below 25,000 dollars, 50% fall between 25,000 and 47,000 dollars and only 25% exceed 47,000 dollars.

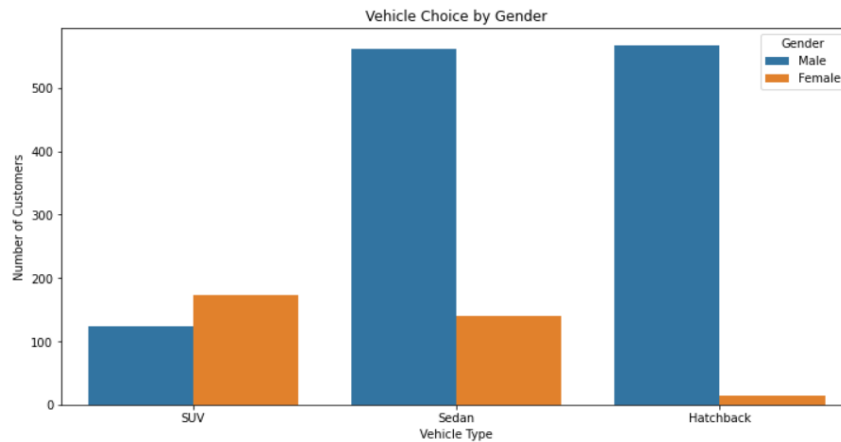
Distribution of Total Household Salary



- The univariate analysis of total household salary indicates a moderately right-skewed distribution.
- Most customers fall within a middle-income bracket (25th–75th percentile: 60,500–95,900), while a smaller segment of higher-income households extends up to 171,000.
- This distribution reflects a diverse customer base, ranging from moderate- to high-income households, and provides a baseline for evaluating affordability and vehicle purchasing patterns.

Bivariate Analysis:

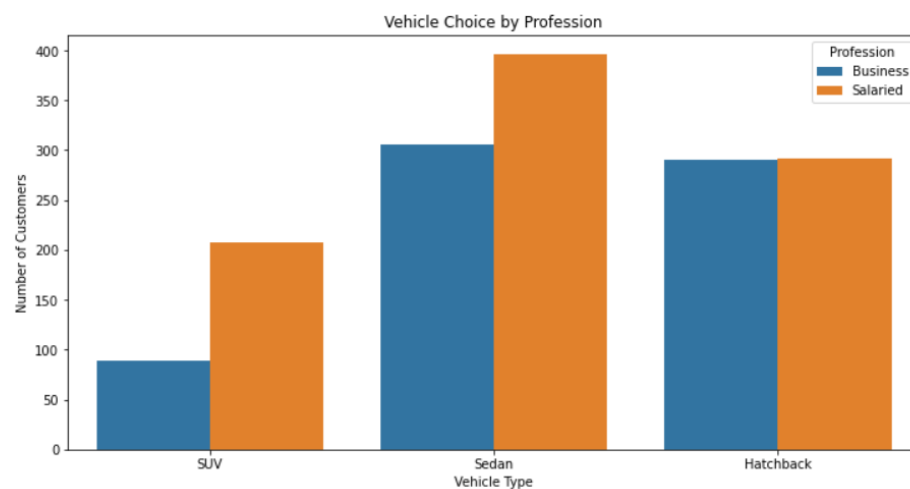
Vehicle Choice by Gender:



Make	Hatchback	SUV	Sedan
Gender			
Female	4.559271	52.583587	42.857143
Male	45.287540	9.904153	44.808307

- The bivariate analysis of Gender vs Vehicle Type reveals a pronounced difference in SUV preference.
- Approximately 53% of female customers purchase SUVs compared to just 10% of male customers.
- Male buyers, in contrast, predominantly prefer Hatchbacks and Sedans.
- These findings indicate a strong gender-specific pattern in vehicle choice, providing an opportunity for Austo Motor Company to tailor marketing campaigns and promotions according to customer gender.

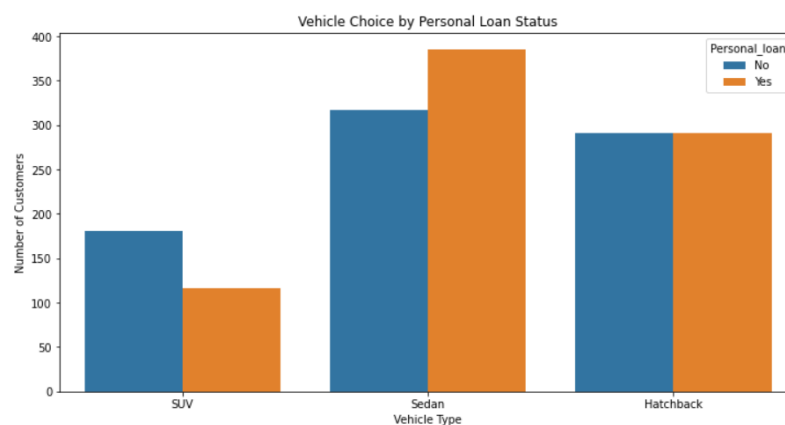
Vehicle Choice by Profession:



Make	Hatchback	SUV	Sedan
Profession			
Business	42.335766	12.992701	44.671533
Salaried	32.589286	23.214286	44.196429

- The bivariate analysis of Profession vs Vehicle Type indicates that Sedans are the most popular choice for both business and salaried customers, accounting for approximately 44% of purchases in each group.
- Salaried individuals are nearly twice as likely as business professionals to purchase SUVs, suggesting that SUV marketing could be targeted toward this segment.
- Conversely, Hatchbacks are more popular among business professionals, reflecting the appeal of practical and cost-efficient options. These insights can guide segment-specific marketing and inventory decisions.

Vehicle Choice by Personal Loan Status:

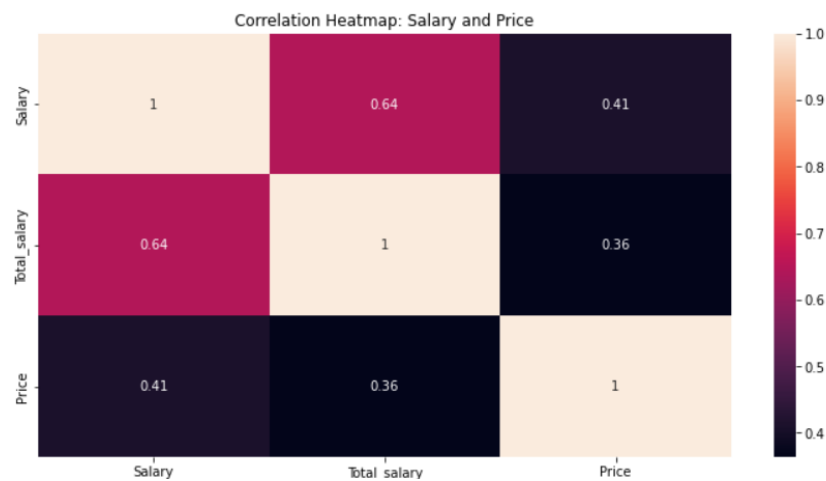
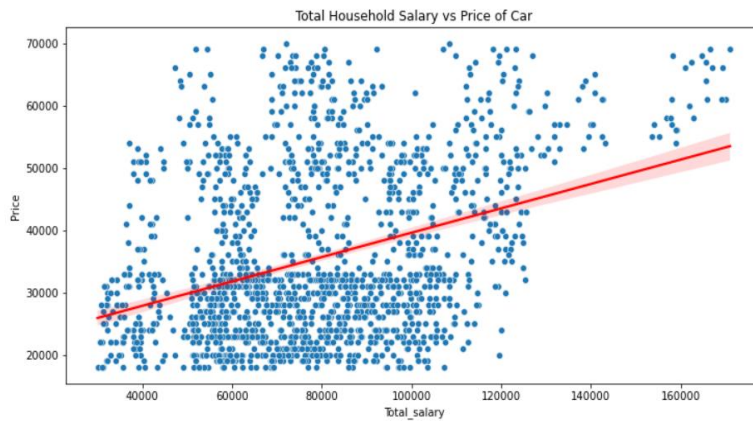


Make	Hatchback	SUV	Sedan
Personal_loan			
No	36.882129	22.940431	40.177440
Yes	36.742424	14.646465	48.611111

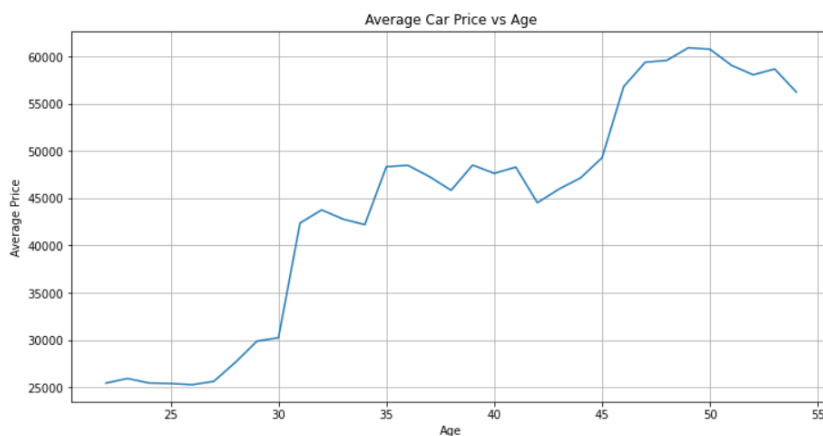
- The bivariate analysis of Personal Loan vs Vehicle Type reveals that customers with personal loans are more likely to purchase Sedans, compared to those without loans.
- SUV purchases are less common among loan users than non-loan customers suggesting that higher-priced SUV adoption may be limited among financed buyers.
- Hatchbacks maintain consistent popularity regardless of loan status, indicating their accessibility across both financed and self-financed segments.
- These insights can inform credit product promotions and vehicle financing strategies.

Price of Car purchase by Salary:

- The bivariate analysis of Salary and Total Household Salary vs Price reveals a moderate positive relationship, with correlations of 0.41 and 0.36 respectively.
- Higher individual and household incomes are associated with purchases of higher-priced vehicles, though the primary buyer's salary appears slightly more predictive than total household income.
- These insights indicate that Austo Motor Company can tailor pricing strategies and marketing efforts according to customer income segments, while also considering other factors that influence purchasing behaviour.

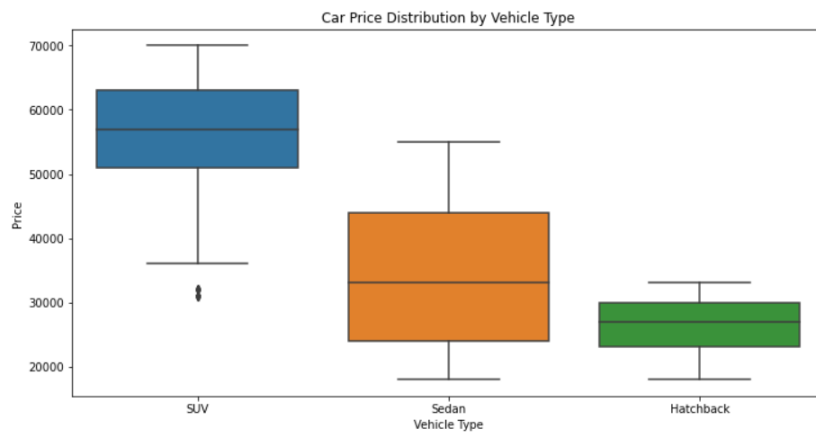


Average Car Price with respect to Age:



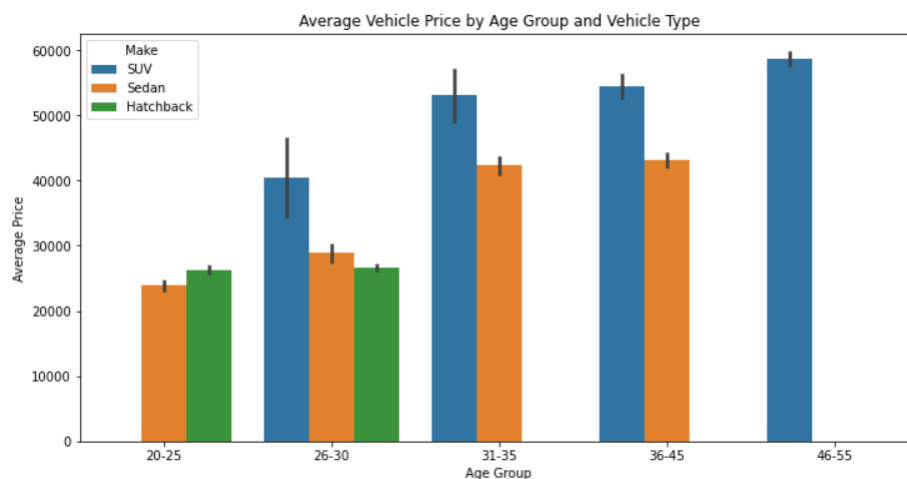
- The line plot analysis of Age vs Car Price indicates a clear positive trend in vehicle spending as customers age.
- Younger customers typically purchase lower-priced cars while middle-aged customers progressively buy higher-priced vehicles, peaking around 45–50 years.
- These insights suggest that Austo Motor Company can tailor pricing and promotional strategies according to age segments, targeting younger customers with entry-level models and middle-aged customers with premium options.

Car Price Distribution by Vehicle Type:



- Hatchbacks occupy the entry-level segment with a median price of 27,000 and limited price variability.
- Sedans form the mid-range category, serving the broadest customer base and exhibiting significant price flexibility.
- SUVs clearly dominate the premium segment, with a median price of 57,000 and a high minimum price threshold, indicating strong willingness to pay among SUV buyers.
- These findings confirm that vehicle type is a primary determinant of price and should remain a key parameter for marketing and pricing strategies at Austo Motor Company.

Average Vehicle Price by Age Group and Vehicle Type:

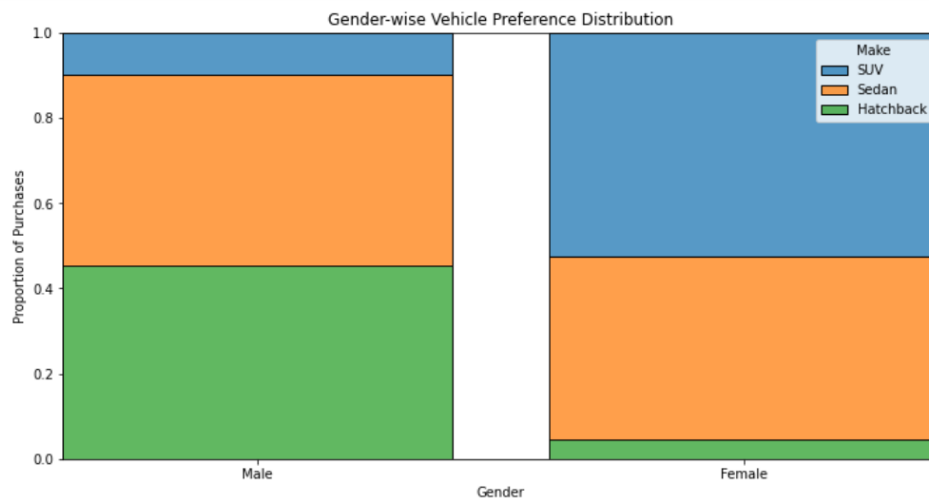


- Vehicle preference and spending are strongly influenced by life stage rather than gender alone.
- Entry-level buyers (20–25) prioritize affordability, while SUV adoption begins in the late 20s and becomes dominant after 30.
- Mid-career customers (36–45) represent the most lucrative segment, consistently purchasing higher-priced SUVs.

Answers to Key Questions:

1] Do men tend to prefer SUVs more compared to women?

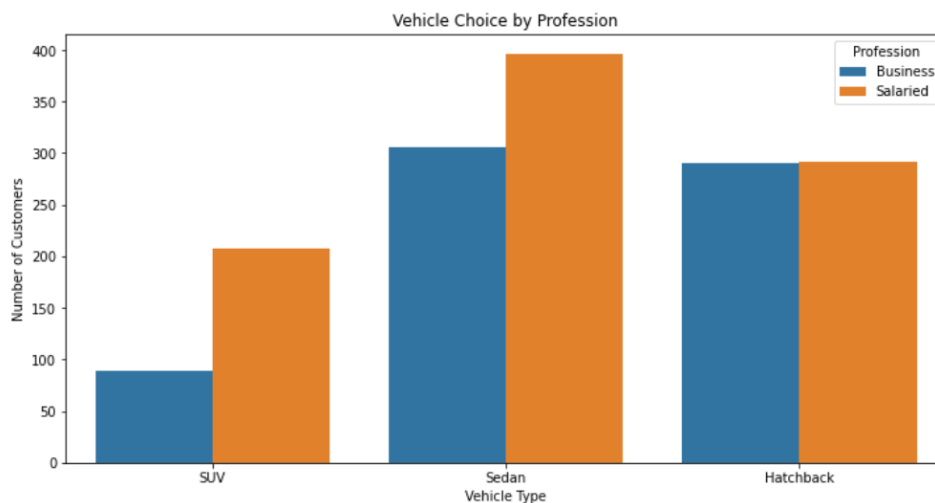
Analysis of gender-wise vehicle preference indicates that men do not prefer SUVs more than women. The SUV preference is significantly higher among female customers. Over 52% of female buyers opted for SUVs, whereas less than 10% of male buyers purchased SUVs. Male customers predominantly preferred hatchbacks and sedans.

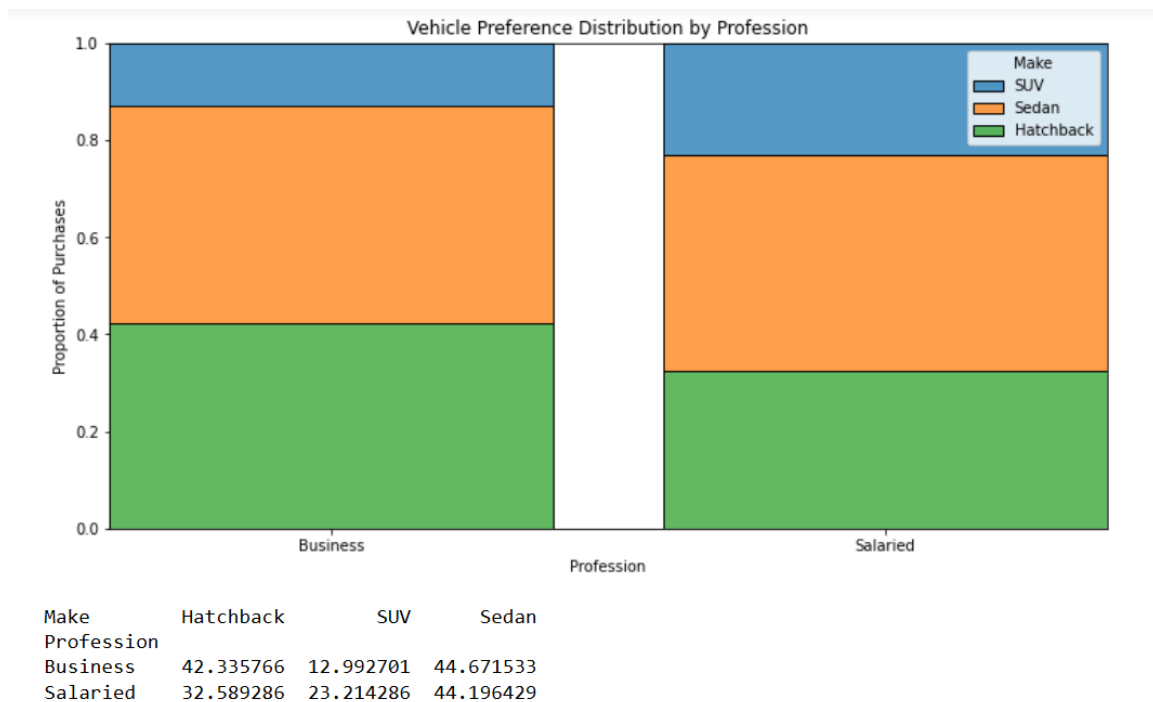


Make	Hatchback	SUV	Sedan
Gender			
Female	4.559271	52.583587	42.857143
Male	45.287540	9.904153	44.808307

2] What is the likelihood of a salaried person buying a Sedan?

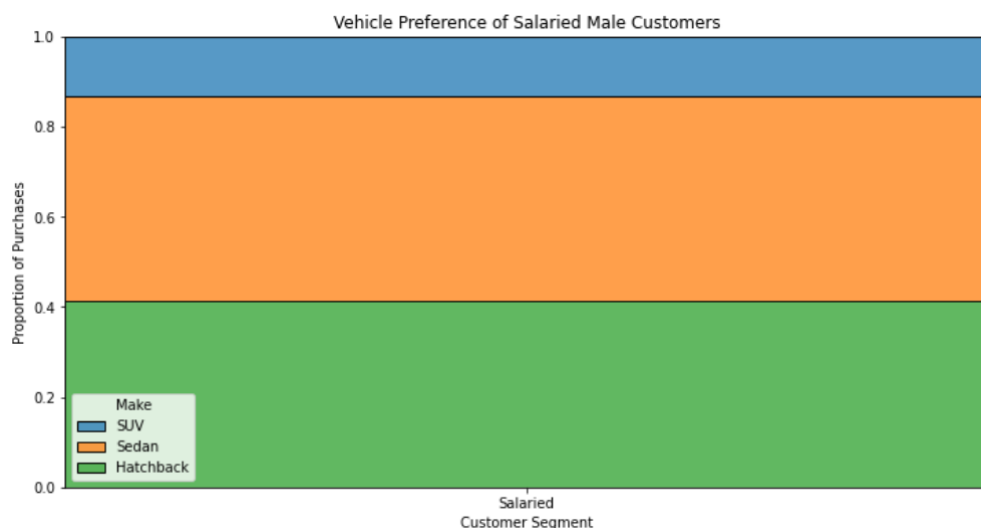
Analysis of profession-wise vehicle preference indicates that salaried individuals show a high likelihood of purchasing Sedans. Approximately 44% of salaried customers opted for Sedans, making it the most preferred vehicle type within this group.

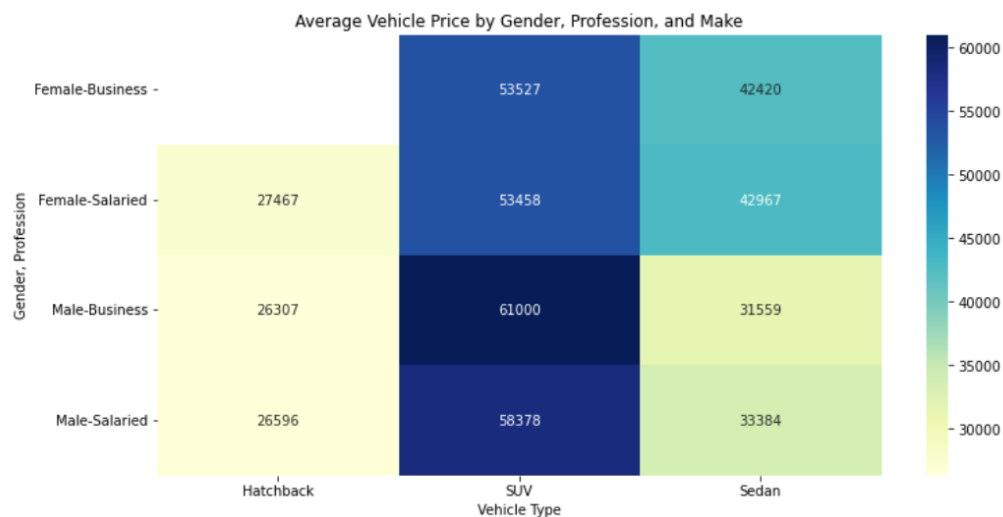




3] What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

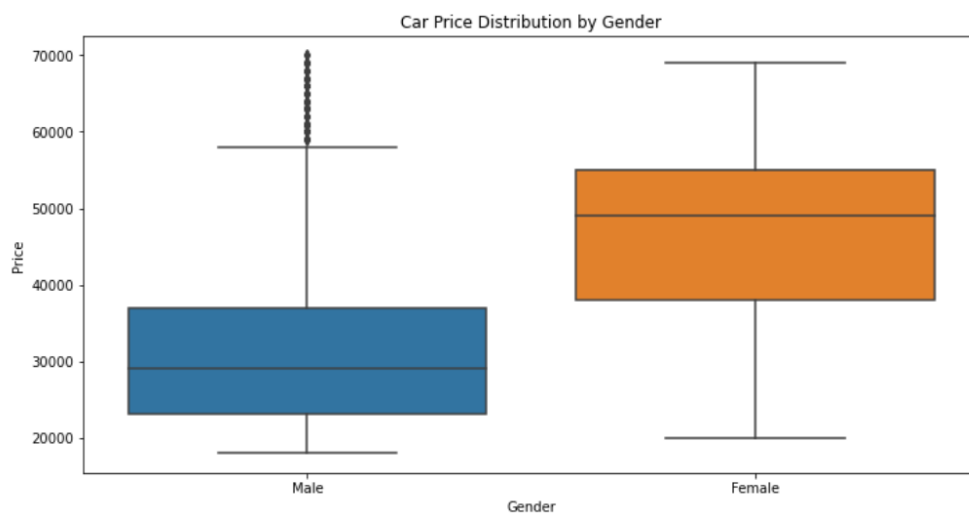
The data provides partial support for Sheldon Cooper's claim that a salaried male is an easier target for an SUV sale compared to a Sedan sale. Within the salaried male segment, while Sedans are still commonly purchased, SUVs are bought at higher prices and generate more value per sale within this group. This indicates that focusing on SUVs for salaried male customers can lead to better revenue outcomes compared to targeting Sedans.





4] How does the amount spent on purchasing automobiles vary by gender?

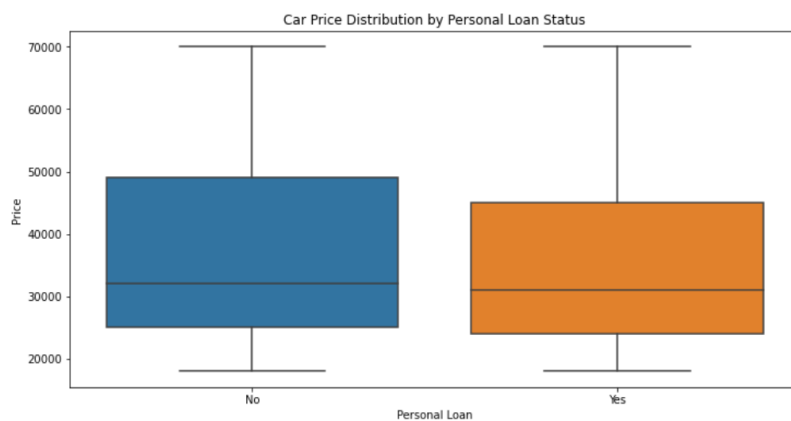
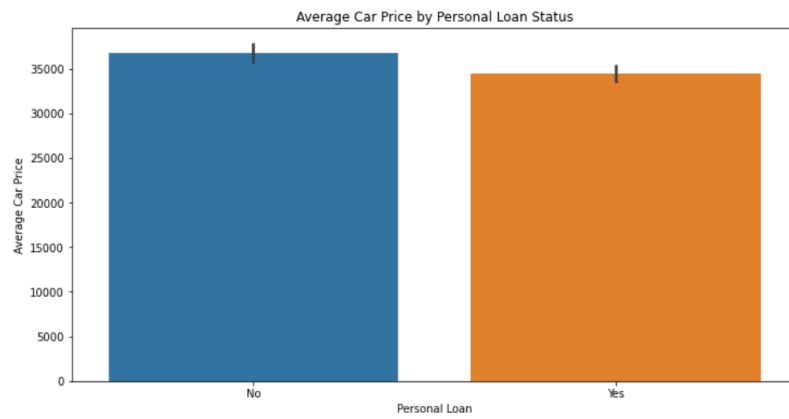
Analysis of automobile purchase prices by gender indicates a significant difference in spending behaviour. Female customers spend considerably more on automobile purchases compared to male customers, with both higher average and median prices. This suggests that female buyers tend to opt for higher-priced vehicles, while male buyers are more concentrated in lower price ranges.



5] How much money was spent on purchasing automobiles by individuals who took a personal loan?

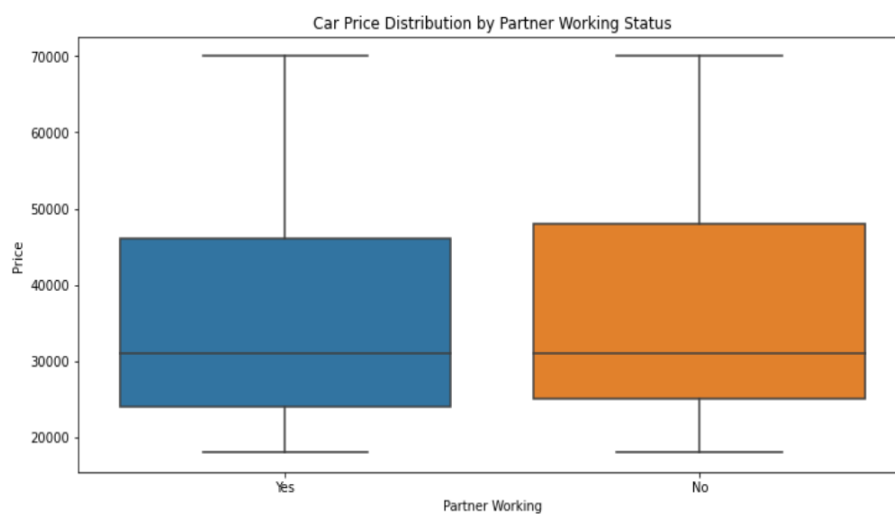
Analysis of automobile purchases based on personal loan status indicates that individuals who availed a personal loan spent approximately 27.2 million dollars on vehicle purchases. Although the average and median purchase prices for personal-loan customers are slightly lower than those for customers without loans, this group contributes a significant share of overall automobile spending.

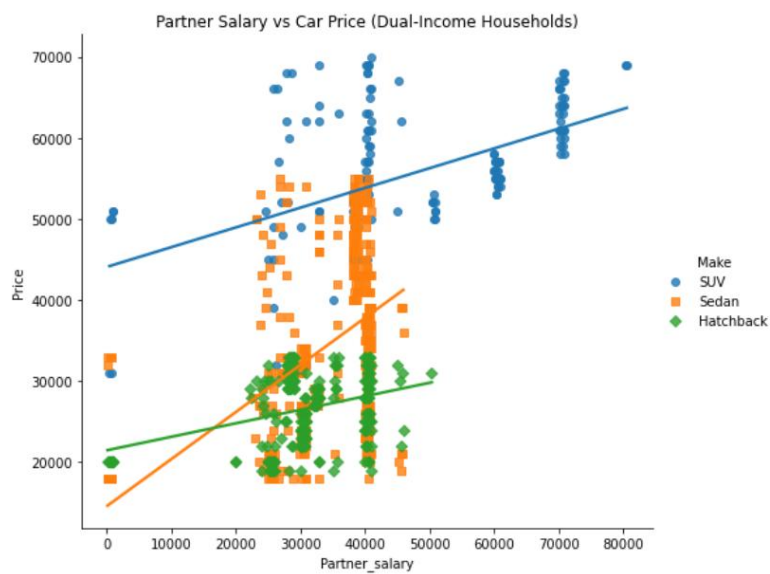
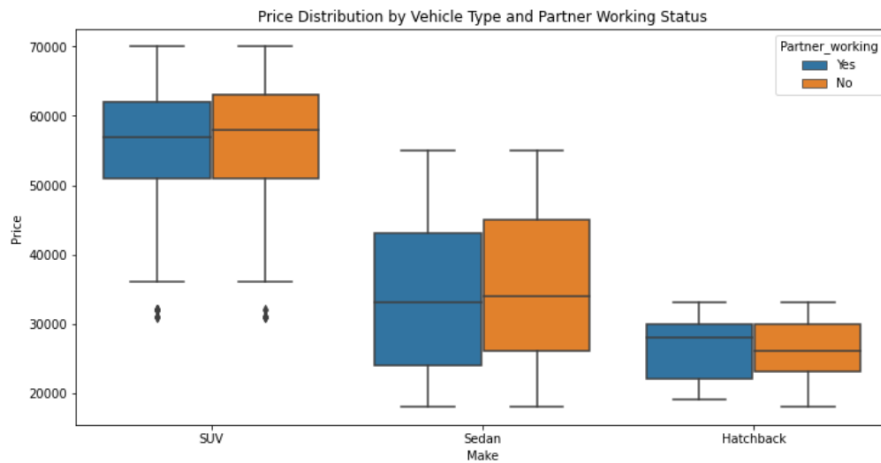
```
Personal_loan
No    28990000
Yes   27290000
Name: Price, dtype: int64
```



6] How does having a working partner influence the purchase of higher-priced cars?

Analysis of automobile purchases indicates that having a working partner does not significantly increase the average or median price of vehicles purchased. Both single-income and dual-income households exhibit similar median spending levels. While partner working status alone does not materially increase average purchase price, higher partner salaries within dual-income households are strongly associated with the purchase of higher-priced vehicle types, particularly SUVs.





```

Partner_working
No    25668000
Yes   30612000
Name: Price, dtype: int64

```

Business Insights:

- Gender plays a major role in vehicle preference, with female customers showing a strong inclination toward SUVs, while male customers predominantly prefer Hatchbacks and Sedans.
- Sedans are the most reliable choice across customer segments, particularly among salaried individuals, where nearly half of purchases fall in this category.
- Salaried male customers represent a high-value opportunity for SUV sales, as SUVs purchased by this group command significantly higher prices compared to Sedans, even if Sedans remain more common in volume.

- Spending behaviour varies significantly by gender, with female customers consistently spending more on automobile purchases than male customers, both on average and at the median level.
- Personal-loan customers contribute significantly to total revenue, accounting for nearly half of overall automobile spending, despite having slightly lower per-vehicle purchase values.
- Dual-income households do not consistently buy more expensive cars, but higher partner salaries within these households are strongly associated with the purchase of premium vehicle types, especially SUVs.
- Vehicle type is the strongest driver of price, with Hatchbacks positioned as entry-level products, Sedans as flexible mid-range options, and SUVs clearly occupying the premium segment.

Recommendations to improve Marketing Strategy:

- Design gender-focused campaigns, positioning SUVs more strongly toward female customers, while promoting Hatchbacks and Sedans as value-driven options for male customers.
- Strengthen Sedan-focused promotions for salaried customers, highlighting comfort, reliability, and affordability, as Sedans already align well with this segment's preferences.
- Target salaried male customers with premium SUV campaigns, emphasizing performance, status, and long-term value, as this group delivers higher revenue per SUV sale.
- Target female buyers with SUV-focused campaigns and premium sedan offers, highlighting safety, comfort, and value-for-money features, since this segment shows a strong preference for higher-priced vehicles.
- Leverage financing-focused marketing for personal-loan customers, bundling Sedans and Hatchbacks with attractive loan offers to maintain volume while improving conversion rates.
- Segment dual-income households by partner salary rather than employment status alone, and target higher partner-income groups with premium SUV offerings.
- Plan inventory and promotions by price range: Use Hatchbacks to drive sales volume, Sedans to maintain steady demand, and SUVs to generate higher profits.